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BUSINESS SERVICE HEALTH SCORE VALIDATION FRAMEWORK

Abstract

Embodiments receive metric and log data from an external system; generate a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models; generate a second set of health scores for log data of the metric and log data using a log data analysis; determine a similarity score based on the first set of health scores and the second set of health scores; and perform a validation by comparing the similarity score to a predetermined threshold value.

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Background/Summary

BACKGROUND

[0001] Aspects of the present invention relate generally to a business service health score validation framework and, more particularly, to a system and method for performing multivariate data analysis to validate business service health scores generated from metrics.

[0002] Business services are a combination of one or more applications. For example, business service health has a topology structure of resource health and application health. In another example, calculating and validating resource health may help in estimating application and business service health.

SUMMARY

[0003] In a first aspect of the invention, there is a computer-implemented method including: receiving, by a computing device, metric and log data from an external system; generating, by the computing device, a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models; generating, by the computing device, a second set of health scores for log data of the metric and log data using a log data analysis; determining, by the computing device, a similarity score based on the first set of health scores and the second set of health scores; and performing, by the computing device, a validation by comparing the similarity score to a predetermined threshold value.

[0004] In another aspect of the invention, there is a computer program product including one or more computer readable storage media having program instructions collectively stored on the one or more computer readable storage media. The program instructions are executable to: receive metric and log data from an external system; generate a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models; generate a second set of health scores for log data of the metric and log data using a log data analysis module; determine a similarity score based on the first set of health scores and the second set of health scores; perform a validation by comparing the similarity score to a predetermined threshold value; and output the first second set of health scores and the second set of health scores and corresponding timestamps to a graphical user interface (GUI).

[0005] In another aspect of the invention, there is system including a processor, a computer readable memory, one or more computer readable storage media, and program instructions collectively stored on the one or more computer readable storage media. The program instructions are executable to: receive metric and log data from an external system; generate a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models which include a matrix profile model, a seasonal trend decomposition (STD) model, an elliptic envelop model, and an isolation forest model; generate a second set of health scores for log data of the metric and log data using a textscalar, a graph embedding component, a graph attention network, a spatial short Fourier transform component, and a gated recurrent unit (GRU); determine a similarity score based on the first set of health scores and the second set of health scores; perform a validation by comparing the similarity score to a predetermined threshold value; and output the first second set of health scores and the second set of health scores and corresponding timestamps to a graphical user interface (GUI). The metric data comprises a central processing unit (CPU) utilization, memory utilization, and network bytes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Aspects of the present invention are described in the detailed description which follows, in reference to the noted plurality of drawings by way of non-limiting examples of exemplary embodiments of the present invention.

[0007] FIG. 1 depicts a cloud computing node according to an embodiment of the present

invention.

[0008] FIG. 2 depicts a cloud computing environment according to an embodiment of the present invention.

[0009] FIG. 3 depicts abstraction model layers according to an embodiment of the present invention.

[0010] FIG. 4 shows a block diagram of a multivariate data analysis system in accordance with aspects of the present invention.

[0011] FIG. 5 shows a block diagram of a metric analysis module in accordance with aspects of the present invention.

[0012] FIG. 6 shows a block diagram of a log data analysis module in accordance with aspects of the present invention.

[0013] FIG. 7 shows another block diagram of the log data analysis module in accordance with aspects of the present invention.

[0014] FIG. 8 shows a validation and similarity module in accordance with aspects of the present invention.

[0015] FIG. 9 shows a flowchart of an exemplary method in accordance with aspects of the present invention.

DETAILED DESCRIPTION

[0016] Aspects of the present invention relate generally to a business service health score validation framework. In more specific embodiments, aspects of the present invention relate to a system, a method, and/or a computer program for performing multivariate data analysis to validate business service health scores generated from metrics. According to aspects of the invention, the system, a method, and/or a computer program is directed to validating a generated health score of a business service from metrics information and improving accuracy of the generated health score. Further, embodiments of the present invention are directed to explaining the generated health score from the metrics information.

[0017] Embodiments of the present invention implement a multi-part framework for identifying anomalies in metrics and log data to generate health scores. Further, implementations of the present invention provide a system, a method, and/or a computer program for validating health scores and comparing the validated health scores to other rule-based functions developed by subject matter experts (SMEs). Aspects of the present invention include an anomaly detection framework which generates a health score from multivariate metrics data.

[0018] Embodiments of the present invention provide an improved understanding of a health impact of a resource, an application, and a business service. Further, aspects of the present invention provide an influence path of each resource, which can help in customer retention and satisfaction. Implementations of the present invention provide a scalable and cost-effective solution for a large and diverse set of customers which have rapid innovation and changes in the marketplace and competitive landscape. Embodiments of the present invention provide an accurate health score which helps in alerting businesses to any persistent issues within a predetermined timeframe (e.g., within the next twenty four hours). Further embodiments of the present invention provide centralized visibility into disparate services and an overall system health. Aspects of the present invention provide predictive health scores which enables pre-emptive and proactive business decisions to ensure there are minimal business incidents. Implementations of the present invention also identify a top health influence of a resource and an application of a business service health and uses the top health influence for future forecasting and explainability. In addition, embodiments of the present invention may be directed to an explainable model stack.

[0019] In more specific embodiments of the present invention, the system, method, and/or computer program provides an artificial intelligence (AI) based multivariate analysis on unstructured data to return a relationship between unstructured log files and their weightage while determining the health score. The system, method, and/or computer program of the present

invention also provides an AI driven health score generator from multivariate unstructured data, which will return the health score from log data sources. Also, implementations of the present invention provide a system, a method, and/or a computer program to return a similarity between captured health scores.

[0020] According to an aspect of the invention, a computer-implemented method includes: measuring an anomaly detection framework which generates a health score from multivariate metrics data; performing an artificial intelligence (AI) based multivariate analysis on unstructured data to return a relationship between unstructured log files and their weightage while determining the health score; an AI driven health score module which generates an AI driven health score from multivariate unstructured data, which will return the health score from log data sources; and returning a similarity between captured health scores.

[0021] Accordingly, implementations of the present invention provide an improvement in the technical field of multivariate data analysis by generating and validating accurate health scores. Generating and validating health scores can be provided by identifying anomalies in metric and log data to generate health scores and validating the generated health scores by determining a similarity score between the generated health scores. In contrast, known systems are not able to directly determine system health scores, analyze multivariate unstructured data, and validate the determined system health scores. Accordingly, known systems have difficulty in determining an application and a health of a system, e.g., memory utilization of the system.

[0022] Implementations of the present invention are necessarily rooted in computer technology. For example, the step of providing an artificial intelligence (AI) based multivariate model on unstructured data to determine a health score is computer-based and cannot be performed in the human mind. Training and building the AI based multivariate model is, by definition, performed by a computer and cannot practically be performed in the human mind (or with pen and paper) due to the complexity and massive amounts of calculations involved. For example, training and building the AI based multivariate model will use a plurality of AI algorithms to determine a health score and validate the health score using similarity scoring, as an example. In particular, training and building the AI based multivariate model performs a large amount of processing of the unstructured data and modeling of parameters to train the AI based multivariate model such that an output of the AI based multivariate model is generated and outputted in real time (or near real time). In other words, the AI based multivariate model is trained using a large amount of previously captured unstructured data and other parameters such that the AI based multivariate model is configured to output health scores in real-time. Given the scale and complexity of processing captured unstructured data and modeling of parameters, it is simply not possible for the human mind, or for a person using pen and paper, to perform the number of calculations involved in training and/or building the AI based multivariate model.

[0023] It should be understood that, to the extent implementations of the invention collect, store, or employ personal information provided by, or obtained from, individuals (for example, users of multivariate metrics data), such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information may be subject to consent of the individual to such activity, for example, through “opt-in” or “opt-out” processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

[0024] The present invention may be a system, a method, and/or a computer program product at any possible technical detail level of integration. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

[0025] The computer readable storage medium can be a tangible device that can retain and store

instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium or media, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

[0026] Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

[0027] Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++, or the like, and procedural programming languages, such as the "C" programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

[0028] Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

[0029] These computer readable program instructions may be provided to a processor of a computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart

and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

[0030] The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0031] The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be accomplished as one step, executed concurrently, substantially concurrently, in a partially or wholly temporally overlapping manner, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

[0032] It is understood in advance that although this disclosure includes a detailed description on cloud computing, implementation of the teachings recited herein are not limited to a cloud computing environment. Rather, embodiments of the present invention are capable of being implemented in conjunction with any other type of computing environment now known or later developed.

[0033] Cloud computing is a model of service delivery for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, network bandwidth, servers, processing, memory, storage, applications, virtual machines, and services) that can be rapidly provisioned and released with minimal management effort or interaction with a provider of the service. This cloud model may include at least five characteristics, at least three service models, and at least four deployment models.

[0034] Characteristics are as follows:

[0035] On-demand self-service: a cloud consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with the service's provider.

[0036] Broad network access: capabilities are available over a network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs).

[0037] Resource pooling: the provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to demand. There is a sense of location independence in that the consumer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter).

[0038] Rapid elasticity: capabilities can be rapidly and elastically provisioned, in some cases automatically, to quickly scale out and rapidly released to quickly scale in. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.

[0039] Measured service: cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported providing transparency for both the provider and consumer of the utilized service.

[0040] Service Models are as follows:

[0041] Software as a Service (SaaS): the capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based e-mail). The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

[0042] Platform as a Service (PaaS): the capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including networks, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.

[0043] Infrastructure as a Service (IaaS): the capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).

[0044] Deployment Models are as follows:

[0045] Private cloud: the cloud infrastructure is operated solely for an organization. It may be managed by the organization or a third party and may exist on-premises or off-premises.

[0046] Community cloud: the cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or a third party and may exist on-premises or off-premises.

[0047] Public cloud: the cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.

[0048] Hybrid cloud: the cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).

[0049] A cloud computing environment is service oriented with a focus on statelessness, low coupling, modularity, and semantic interoperability. At the heart of cloud computing is an infrastructure comprising a network of interconnected nodes.

[0050] Referring now to FIG. 1, a schematic of an example of a cloud computing node is shown. Cloud computing node **10** is only one example of a suitable cloud computing node and is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the invention described herein. Regardless, cloud computing node **10** is capable of being implemented and/or performing any of the functionality set forth hereinabove.

[0051] In cloud computing node **10** there is a computer system/server **12**, which is operational with

numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system/server **12** include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

[0052] Computer system/server **12** may be described in the general context of computer system executable instructions, such as program modules, being executed by a computer system.

Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types.

Computer system/server **12** may be practiced in distributed cloud computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

[0053] As shown in FIG. **1**, computer system/server **12** in cloud computing node **10** is shown in the form of a general-purpose computing device. The components of computer system/server **12** may include, but are not limited to, one or more processors or processing units **16**, a system memory **28**, and a bus **18** that couples various system components including system memory **28** to processor **16**.

[0054] Bus **18** represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnects (PCI) bus.

[0055] Computer system/server **12** typically includes a variety of computer system readable media. Such media may be any available media that is accessible by computer system/server **12**, and it includes both volatile and non-volatile media, removable and non-removable media.

[0056] System memory **28** can include computer system readable media in the form of volatile memory, such as random access memory (RAM) **30** and/or cache memory **32**. Computer system/server **12** may further include other removable/non-removable, volatile/non-volatile computer system storage media. By way of example only, storage system **34** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). Although not shown, a magnetic disk drive for reading from and writing to a removable, non-volatile magnetic disk (e.g., a “floppy disk”), and an optical disk drive for reading from or writing to a removable, non-volatile optical disk such as a CD-ROM, DVD-ROM or other optical media can be provided. In such instances, each can be connected to bus **18** by one or more data media interfaces. As will be further depicted and described below, memory **28** may include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out the functions of embodiments of the invention.

[0057] Program/utility **40**, having a set (at least one) of program modules **42**, may be stored in memory **28** by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. Program modules **42** generally carry out the functions and/or methodologies of embodiments of the invention as described herein.

[0058] Computer system/server **12** may also communicate with one or more external devices **14** such as a keyboard, a pointing device, a display **24**, etc.; one or more devices that enable a user to

interact with computer system/server **12**; and/or any devices (e.g., network card, modem, etc.) that enable computer system/server **12** to communicate with one or more other computing devices. Such communication can occur via Input/Output (I/O) interfaces **22**. Still yet, computer system/server **12** can communicate with one or more networks such as a local area network (LAN), a general wide area network (WAN), and/or a public network (e.g., the Internet) via network adapter **20**. As depicted, network adapter **20** communicates with the other components of computer system/server **12** via bus **18**. It should be understood that although not shown, other hardware and/or software components could be used in conjunction with computer system/server **12**.

Examples, include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, tape drives, and data archival storage systems, etc. [0059] Referring now to FIG. **2**, illustrative cloud computing environment **50** is depicted. As shown, cloud computing environment **50** comprises one or more cloud computing nodes **10** with which local computing devices used by cloud consumers, such as, for example, personal digital assistant (PDA) or cellular telephone **54A**, desktop computer **54B**, laptop computer **54C**, and/or automobile computer system **54N** may communicate. Nodes **10** may communicate with one another. They may be grouped (not shown) physically or virtually, in one or more networks, such as Private, Community, Public, or Hybrid clouds as described hereinabove, or a combination thereof. This allows cloud computing environment **50** to offer infrastructure, platforms and/or software as services for which a cloud consumer does not need to maintain resources on a local computing device. It is understood that the types of computing devices **54A-N** shown in FIG. **2** are intended to be illustrative only and that computing nodes **10** and cloud computing environment **50** can communicate with any type of computerized device over any type of network and/or network addressable connection (e.g., using a web browser).

[0060] Referring now to FIG. **3**, a set of functional abstraction layers provided by cloud computing environment **50** (FIG. **2**) is shown. It should be understood in advance that the components, layers, and functions shown in FIG. **3** are intended to be illustrative only and embodiments of the invention are not limited thereto. As depicted, the following layers and corresponding functions are provided:

[0061] Hardware and software layer **60** includes hardware and software components. Examples of hardware components include: mainframes **61**; RISC (Reduced Instruction Set Computer) architecture based servers **62**; servers **63**; blade servers **64**; storage devices **65**; and networks and networking components **66**. In some embodiments, software components include network application server software **67** and database software **68**.

[0062] Virtualization layer **70** provides an abstraction layer from which the following examples of virtual entities may be provided: virtual servers **71**; virtual storage **72**; virtual networks **73**, including virtual private networks; virtual applications and operating systems **74**; and virtual clients **75**.

[0063] In one example, management layer **80** may provide the functions described below. Resource provisioning **81** provides dynamic procurement of computing resources and other resources that are utilized to perform tasks within the cloud computing environment. Metering and Pricing **82** provide cost tracking as resources are utilized within the cloud computing environment, and billing or invoicing for consumption of these resources. In one example, these resources may comprise application software licenses. Security provides identity verification for cloud consumers and tasks, as well as protection for data and other resources. User portal **83** provides access to the cloud computing environment for consumers and system administrators. Service level management **84** provides cloud computing resource allocation and management such that required service levels are met. Service Level Agreement (SLA) planning and fulfillment **85** provide pre-arrangement for, and procurement of, cloud computing resources for which a future requirement is anticipated in accordance with an SLA.

[0064] Workloads layer **90** provides examples of functionality for which the cloud computing

environment may be utilized. Examples of workloads and functions which may be provided from this layer include: mapping and navigation **91**; software development and lifecycle management **92**; virtual classroom education delivery **93**; data analytics processing **94**; transaction processing **95**; and multivariate data analysis **96**.

[0065] Implementations of the invention may include a computer system/server **12** of FIG. **1** in which one or more of the program modules **42** are configured to perform (or cause the computer system/server **12** to perform) one of more functions of the multivariate data analysis **96** of FIG. **3**. For example, the one or more of the program modules **42** may be configured to: receive metric and log data from an external system; generate a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models which include a matrix profile model, a seasonal trend decomposition (STD) model, an elliptic envelop model, and an isolation forest model; generate a second set of health scores for log data of the metric and log data using a textscalar, a graph embedding component, a graph attention network, a spatial short Fourier transform component, and a gated recurrent unit (GRU); determine a similarity score based on the first set of health scores and the second set of health scores; and perform a validation by comparing the similarity score to a predetermined threshold value.

[0066] FIG. **4** shows a block diagram of an attack detection system in accordance with aspects of the invention. In embodiments, the multivariate data analysis system **100** comprises a multivariate data analysis environment **105** which includes a multi-part framework module **110**, a metric analysis module **115**, a log data analysis module **120**, and a validation and similarity module **130**, each of which may comprise one or more program modules such as program modules **42** described with respect to FIG. **1** and the multivariate data analysis **96** of FIG. **3**.

[0067] The multivariate data analysis system **100** may include additional or fewer modules than those shown in FIG. **4**. In embodiments, separate modules may be integrated into a single module. Additionally, or alternatively, a single module may be implemented as multiple modules. Moreover, the quantity of devices and/or networks in the environment is not limited to what is shown in FIG. **4**. In practice, the environment may include additional devices and/or networks; fewer devices and/or networks; different devices and/or networks; or differently arranged devices and/or networks than illustrated in FIG. **4**.

[0068] In embodiments of FIG. **4**, the multi-part framework module **110** receives metric and log data from an external system. In embodiments, the multi-part framework module **110** comprises a metric analysis module **115** which performs AI based multivariate analysis on unstructured metric data to determine a first set of health scores. In further embodiments, the multi-part framework module **110** comprises a log data analysis module **120** which performs AI based multivariate analysis on unstructured log data to determine a second set of health scores. In embodiments, the multi-part framework module **110** outputs the first set of health scores and the second set of health scores to the validation and similarity module **130**. Details of the metric analysis module **115** and the log data analysis module **120** are provided in FIGS. **5-7**.

[0069] In embodiments of FIG. **4**, the validation and similarity module **130** receives the first set of health scores and the second set of health scores. In embodiments, the validation and similarity module **130** validates the first and second set of health scores and outputs a similarity score between the first and second set of health scores. In further embodiments, the validation and similarity module **130** also receives a third set of health scores from rule-based functions developed by subject matter experts (SMEs). In this scenario, the validating and similarity module **130** validates the first and third set of health scores and outputs a similarity score between the first and third set of health scores. Further, the validating and similarity module **130** validates the second and third set of health scores and outputs a similarity score between the second and third set of health scores. Moreover, the validating and similarity module **130** validates the first and second set of health scores and outputs a similarity score between the first and second set of health scores. In embodiments, the validation and similarity module **130** outputs a text summarization explainability

message **385** to an external system (see FIG. **8**). Details of the validation and similarity module **130** are provided in FIG. **8**.

[0070] FIG. **5** shows a block diagram of a metric analysis module in accordance with aspects of the present invention. In embodiments, the metric analysis module **115** receives the unstructured metric data **135** and then inputs the unstructured metric data to an anomaly detection framework (ADF) **140**. In embodiments, the unstructured metric data **135** comprises unstructured multivariate metric data such as a central processing unit (CPU) utilization, memory utilization, network bytes, etc. The ADF **140** comprises a unified framework which is utilized for an application of a computing device. In further embodiments, the ADF **140** comprises a plurality of AI models including a matrix profile model **145**, a seasonal trend decomposition (STD) model **150**, an elliptic envelop model **155**, and isolation forest model **160**.

[0071] In embodiments of FIG. **5**, the matrix profile model **145** of the ADF **140** receives the unstructured metric data **135** and calculates anomaly points and anomaly scores using a matrix profile algorithm which computes distances for a windowed sub-sequence against an entire time series, sets an exclusion zone to ignore trivial matches, updates a distance profile with minimal values, and sets a first nearest-neighbor index. In further embodiments, the STD model **150** of the ADF **140** receives the unstructured metric data **135** and calculates anomaly points and anomaly scores using a STD algorithm, which uses locally fitted regression models to decompose a time series into trend, seasonal, and remainder components. The elliptic envelop model **155** of the ADF **140** receives the unstructured metric data **135** and calculates anomaly points and anomaly scores using an elliptic envelop algorithm, which creates an elliptical area around a dataset of the unstructured metric data **135**. In further implementations of the present invention, the isolation forest model **160** of the ADF **140** receives the unstructured metric data **135** and calculates anomaly points and anomaly scores using an isolation forest algorithm, which splits a data space randomly and uses a randomly selected attribute and a randomly selected split point to detect anomalies. The ADF **140** will output an intersection of all anomaly points **165** and corresponding anomaly scores **170** from the matrix profile model **145**, the STD model **150**, the elliptic envelop model **155**, and the isolation forest model **160**.

[0072] In further embodiments of FIG. **5**, the corresponding anomaly scores **170** are normalized **175**. After normalization **175**, the normalized anomaly scores are used in a health score generation **180** to generate health scores **185**. Accordingly, the health score generation **180** outputs the health scores **185** of the health of a business service.

[0073] FIG. **6** shows a block diagram of a log data analysis module in accordance with aspects of the present invention. In embodiments, the log data analysis module **120** receives the unstructured log data **190** and sends the unstructured log data **190** to a textscalar **195**. In embodiments, the textscalar **195** converts each log at a specific timestamp of the unstructured log data **190** to a scalar value and multiplies the converted scalar value to create row vectors per log file. In embodiments, the row vectors per log file are sent to a graph embedding component **200**. The graph embedding component **200** receives the row vectors per log file and uses a neural network (NN) algorithm to return graph data which includes links between log files based on their Euclidean distance. In embodiments, the graph embedding component **200** sends the graph data comprising the links between log files based on their Euclidean distance to a graph neural network (GNN) **205**. The GNN **205** receives the graph data and generates node embedding vectors which comprise vectors that represent the Euclidean distance and other similarity characteristics between the nodes in the graph data. The GNN **205** sends the generated node embedding vectors to an eigenvector centrality component **210** and a graph attention network (GAT) **220**.

[0074] In embodiments of FIG. **6**, the eigenvector centrality component **210** receives the generated node embedding vectors and determines a weightage of each log file **215** by using an eigenvector centrality algorithm to measure a transitive influence of nodes based on relationships originating from high-scoring nodes which contribute more to the score of a node than connections from low-

scoring nodes. In further embodiments, the GAT **220** finds a relationship between the log files by passing two node embedding vectors (i.e., $Wx_{sub.i}$, $Wx_{sub.j}$) at a time to output a weight attention score **225**. In embodiments, the weight attention score **225** comprises a relationship between nodes. The GAT **220** performs the above operation for all possible combination of nodes. [0075] FIG. 7 shows another block diagram of the log data analysis module in accordance with aspects of the present invention. In embodiments, the log data analysis module **120** receives the unstructured log data **190** and sends the unstructured log data **190** to the textscaler **195**. In embodiments, the textscaler **195** converts each log at a specific timestamp of the unstructured log data **190** to a scalar value and multiplies the converted scalar value to create row vectors per log file. The textscaler **195** converts the row vectors per log file into a matrix. The textscaler **195** sends the matrix to a spatial short Fourier transform component **230** and another graph attention network (GAT) **240**.

[0076] In embodiments of FIG. 7, the spatial short Fourier transform component **230** receives the matrix and converts the matrix into a spatial dimension and temporal dimension of a lower space **235**. The GAT **240** will receive the weightage of each log file **215** and the weight attention score **225** from a multivariate analysis **245** using the log data analysis module **120** of FIG. 6. As described above in FIG. 6, the weight attention score **225** comprises the relationship between nodes. The GAT **240** will output a matrix **260** based the weightage of each log file **215**, the weight attention score **225**, and the matrix from the textscaler **195** using learning attention functions that assign weights to nodes in a graph and allow different nodes to have varying influences during the feature aggregation process. The GAT **240** sends the matrix **260** to be concatenated with the spatial dimension and temporal dimension of the lower space **235** at a concatenation operator module **263**. The concatenated matrix is sent by the concatenation operator module **263** to a gated recurrent unit (GRU) **275**.

[0077] In embodiments of FIG. 7, the GRU **275** comprises a gating mechanism in recurrent neural networks to input or forget certain features. In particular, the GRU **275** outputs the concatenated matrix which comprises a concatenation of the matrix **260** and the spatial dimension and temporal dimension of the lower space **235** to a forecasting based model **265**. The forecasting model **265** returns forecasted anomaly scores and averages the forecasted anomaly scores with a huber loss **280**. The forecasting based model **265** sends the averaged forecasted anomaly scores after the huber loss to the anomaly scores component **290**. The GRU **275** also outputs the concatenated matrix to a reconstruction based model **270**. The reconstruction based model **270** returns reconstructed anomaly scores and averages the reconstructed anomaly scores with another huber loss **285**. In further embodiments, the reconstruction based model **270** receives the concatenated matrix and assigns a reconstructed anomaly score to each object in the concatenated matrix based on a similarity of each object to an overall data in the concatenated matrix. The reconstruction based model **270** sends the averaged reconstructed anomaly scores after the huber loss to the anomaly scores component **290**.

[0078] In further embodiments of FIG. 7, the anomaly scores component **290** generates final anomaly scores based on the average of the reconstructed anomaly scores after the huber loss and the averaged forecasted anomaly scores after the huber loss. In embodiments, the huber loss comprises a loss function which makes the average of the reconstructed anomaly scores and the average forecasted anomaly scores less sensitive to outlier anomalies and minimizes errors within the average of the reconstructed anomaly scores and the average forecasted anomaly scores. Then, the generated final anomaly scores are normalized **300**. In embodiments, the generated final anomaly scores are normalized **300** by scaling the generated final anomaly scores in a range of 0 to 100, in which **100** represents a highest anomaly score and 0 represents a lowest anomaly score. After the normalization **300**, the normalized final anomaly scores are used in a health score generation **305** to generate health scores **310** (as shown in FIG. 8). Accordingly, the health score generation **305** outputs the health scores **310** of the business service health.

[0079] FIG. 8 shows a validation and similarity module in accordance with aspects of the present invention. In embodiments, the validation and similarity module **130** includes a system health module **320** which receives the health scores **185** corresponding to the unstructured metric data **135**, the health scores **310** corresponding to the unstructured log data **190**, and health scores **325** from rule-based functions. In aspects of the present invention, the system health module **320** also receives timestamps corresponding to the health scores **185**, **310**, and **325**. In further embodiments, the health scores **325** may be determined based on rule-based functions developed by SMEs; however, embodiments are not limited to this example. For example, the system health **320** may only receive the health scores **185** corresponding to the unstructured metric data **135** and the health scores **310** corresponding to the unstructured log data **190**.

[0080] In embodiments of FIG. 8, a time wrapped edit distance (TWED) component **330** measures a similarity between the health scores **185** and the health scores **310**, a similarity between the health scores **185** and the health scores **325**, and a similarity between the health scores **310** and the health scores **325** using a discrete time series matching algorithm with time elasticity. In particular, the time wrapped edit distance component **300** uses the discrete time series matching algorithm with time elasticity to quantify how similar the health scores are to each other by counting a minimum number of operations required to transform one of the health scores to the other health score. In aspects of the present invention, the time wrapped edit distance (TWED) component **330** outputs the similarity between the health scores **185** and the health scores **310**, the similarity between the health scores **185** and the health scores **325**, and the similarity between the health scores **310** and the health scores **325** to a sigmoid activation function **340**. The sigmoid activation function **340** receives a bias input **335** from a deep learning model and outputs a first similarity score (FSS) **345** corresponding to the similarity between the health scores **185** and the health score **325**, a second similarity score (SSS) **350** corresponding to the similarity between the health scores **185** and the health scores **325**, and a third similarity score (TSS) **355** corresponding to the similarity between the health scores **310** and the health scores **325** to an average similarity score (ASC) component **360**. The bias input **335** is a constant value from the deep learning model that allows a shifting of the sigmoid activation function **340** towards predetermined output values.

[0081] In further embodiments of FIG. 8, the first similarity score (FSS) **345**, the second similarity score (SSS) **350**, and the third similarity score (TSS) **355** are a numerical value between a range of zero (e.g., no similarity) and one (absolute match in similarity). The average similarity score component (ASC) **360** then averages the first similarity score (FSS) **345**, the second similarity score (SSS) **350**, and the third similarity score (TSS) **355** together and outputs an averaged similarity score to a threshold component **365**. In embodiments, the threshold component **365** compares the averaged similarity score to a predetermined threshold value. The threshold component **365** outputs a validation failed message **370** in response to the average similarity score being less than the predetermined threshold value and sends the average similarity score and the validation failed message to a text summarization component **380**.

[0082] Still referring to FIG. 8, the threshold component **365** outputs a validation success message **375** in response to the average similarity score being greater than the predetermined threshold value and sends the average similarity score and the validation successful message to the text summarization component **380**. The text summarization component **380** receives the average similarity score, one of the validation failed message **370** and the validation success message **375**, the health scores **185**, **310**, and **325** and corresponding timestamps from the system health module **320**. The text summarization component **380** outputs a text summarization explainability message **385** to a graphical user input (GUI) of a display which includes values of the average similarity score, the health scores **185**, **310**, and **325** and corresponding timestamps, and a validation message which includes one of the validation failed message **370** and the validation success message **375**.

[0083] FIG. 9 shows a flowchart of an exemplary method in accordance with aspects of the present invention. Steps of the method may be carried out in the environment of FIGS. 4-8 and are

described with reference to elements depicted in FIGS. 4-8.

[0084] At step **905**, the system receives, at the multi-framework module **110**, metric and log data from an external system. In embodiments, and as described with respect to FIG. 4, the multi-framework module **110** further comprises a metric analysis module **115** and a log data analysis module **120**.

[0085] At step **910**, the system generates, at the metric analysis module **115**, health scores **185** corresponding to unstructured metric data **135**. In embodiments, and as described with respect to FIGS. 4-8, the metric analysis module **115** generates the health scores **185** using an anomaly detection framework (ADF) **140** which comprises a plurality of AI models including a matrix profile model **145**, a seasonal trend decomposition (STD) model **150**, an elliptic envelop model **155**, and an isolation forest model **160**. In an example, a health score represents a health of a system, e.g., memory utilization of the system.

[0086] At step **915**, the system generates, at the log data analysis module **120**, health scores **310** corresponding to unstructured log data **190**. In embodiments, and as described with respect to FIGS. 4-8, the log data analysis module **120** generates the health scores **310** using a textscalar **195**, a graph embedding component **200**, at least one of graph attention networks (GATs) **220**, **240**, a spatial short Fourier transform component **230**, and a gated recurrent unit (GRU) **275**.

[0087] At step **920**, the system performs, at the validation and similarity module **130**, a validation based on the health scores **185** corresponding to unstructured metric data **135**, the health scores **310** corresponding to the unstructured log data **190**, and the health scores **325** corresponding to rule-based functions developed by SMEs. In embodiments, and as described with respect to FIGS. 4-8, the validation and similarity module **130** also determines a similarity score between the health scores **185**, the health scores **310**, and the health scores **325**. In further embodiments, and as described with respect to FIG. 4-8, the validation and similarity module **130** performs the validation by comparing the similarity score with a predetermined threshold value.

[0088] At step **925**, the system outputs, at the validation and similarity module **130**, a text summarization explainability message **385** to a graphical user input (GUI) of a display. In embodiments, and as described with respect to FIGS. 4-8, the validation and similarity module **130** outputs the text summarization explainability message **385** which includes values of the similarity scores, the health scores **185**, **310**, **325** and corresponding timestamps, and a validation message which includes one of the validation failed message **370** and the validation success message **375** in response to a determination of whether the validation was successful.

[0089] In embodiments, a service provider could offer to perform the processes described herein. In this case, the service provider can create, maintain, deploy, support, etc., the computer infrastructure that performs the process steps of the invention for one or more customers. These customers may be, for example, any business that uses technology. In return, the service provider can receive payment from the customer(s) under a subscription and/or fee agreement and/or the service provider can receive payment from the sale of advertising content to one or more third parties.

[0090] In still additional embodiments, the invention provides a computer-implemented method, via a network. In this case, a computer infrastructure, such as computer system/server **12** (FIG. 1), can be provided and one or more systems for performing the processes of the invention can be obtained (e.g., created, purchased, used, modified, etc.) and deployed to the computer infrastructure. To this extent, the deployment of a system can comprise one or more of: (1) installing program code on a computing device, such as computer system/server **12** (as shown in FIG. 1), from a computer-readable medium; (2) adding one or more computing devices to the computer infrastructure; and (3) incorporating and/or modifying one or more existing systems of the computer infrastructure to enable the computer infrastructure to perform the processes of the invention.

[0091] The descriptions of the various embodiments of the present invention have been presented

for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A method, comprising: receiving, by a computing device, metric and log data from an external system; generating, by the computing device, a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models; generating, by the computing device, a second set of health scores for log data of the metric and log data using a log data analysis; determining, by the computing device, a similarity score based on the first set of health scores and the second set of health scores; and performing, by the computing device, a validation by comparing the similarity score to a predetermined threshold value.
2. The method of claim 1, wherein the metric data comprises at least one of a central processing unit (CPU) utilization, memory utilization, and network bytes.
3. The method of claim 1, further comprising outputting, by the computing device, a validation failed message in response to the similarity score being less than the predetermined threshold value.
4. The method of claim 1, further comprising outputting, by the computing device, a validation successful message in response to the similarity score being greater than the predetermined threshold value.
5. The method of claim 1, wherein the determining the similarity score based on the first set of health scores and the second set of health scores occurs by using a time wrapped edit distance component between the first set of health scores and the second set of health scores.
6. The method of claim 1, wherein the matrix profile model receives the metric data and calculates anomaly points and anomaly scores using a matrix profile algorithm which computes distances for a windowed sub-sequence against an entire time series, sets an exclusion zone to ignore trivial matches, updates a distance profile with minimal values, and sets a first nearest-neighbor index.
7. The method of claim 1, wherein the seasonal trend decomposition (STD) model receives the metric data and calculates anomaly points and anomaly scores using a STD algorithm which uses locally fitted regression models to decompose a time series into trend, seasonal, and remainder components.
8. The method of claim 1, wherein the elliptic envelop model receives the metric data and calculates anomaly points and anomaly scores using an elliptic envelop algorithm which creates an elliptical area around a dataset of the metric data.
9. The method of claim 1, wherein the isolation forest model receives the metric data and calculates anomaly points and anomaly scores using an isolation forest algorithm which splits a data space randomly and uses a randomly selected attribute and a randomly selected split point to detect anomalies.
10. The method of claim 1, further comprising: generating, by the computing device, a third set of health scores from rule based functions developed by subject matter expert (SME) rules; determining, by the computing device, a second similarity score based on the first set of health scores and the third set of health scores; determining, by the computing device, a third similarity score based on the second set of health scores and the third set of health scores; averaging, by the computing device, the similarity score, the second similarity score, and the third similarity score together to create an average similarity score; and validating, by the computing device, the average similarity score by comparing the average similarity score to the predetermined threshold value.

11. The method of claim 1, further comprising outputting, by the computing device, the first set of health scores and the second set of health scores and corresponding timestamps.

12. The method of claim 1, wherein the plurality of AI models comprise a matrix profile model, a seasonal trend decomposition (STD) model, an elliptic envelope model, and an isolation forest model.

13. The method of claim 12, wherein the matrix profile model receives the metric data and calculates a first set of anomaly points and anomaly scores using a matrix profile algorithm which computes distances for a windowed sub-sequence against an entire time series, sets an exclusion zone to ignore trivial matches, and updates a distance profile with minimal values, and sets a nearest-neighbor index, and the STD model receives the metric data and calculates a second set of anomaly points and anomaly scores using a STD algorithm which uses locally fitted regression models to decompose a time series into trend, seasonal, and remainder components.

14. The method of claim 12, wherein the elliptic envelop model receives the metric data and calculates a third set of anomaly points and anomaly scores using an elliptic envelop algorithm which creates an elliptical area around a dataset of the metric data, and the isolation forest model receives the metric data and calculates a fourth set of anomaly points and anomaly scores using an isolation forest algorithm which splits a data space randomly and uses a randomly selected attribute and a randomly selected split point to detect anomalies.

15. The method of claim 1, wherein the log data analysis comprises: a textscalar which converts each log at a specific timestamp of the log data to a scalar value and multiplies the converted scalar value to create row vectors per file; a graph embedding component which receives the row vectors per file and uses a neural network (NN) algorithm to return links between log files based on their Euclidean distance; a graph attention network which receives graph data comprising the links between the log files based on their Euclidean distance and generates node embedding vectors; a spatial short Fourier transform component which receives a matrix based on the row vector per file and converts the matrix into a spatial dimension and temporal dimension of a lower space; and a gated recurrent unit (GRU) which outputs a concatenated matrix which is a concatenation of the spatial dimension and temporal dimension of the lower space and another matrix based on a weightage of the log file.

16. A computer program product comprising one or more computer readable storage media having program instructions collectively stored on the one or more computer readable storage media, the program instructions executable to: receive metric and log data from an external system; generate a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models; generate a second set of health scores for log data of the metric and log data using a log data analysis module; determine a similarity score based on the first set of health scores and the second set of health scores; perform a validation by comparing the similarity score to a predetermined threshold value; and output the first set of health scores and the second set of health scores and corresponding timestamps to a graphical user interface (GUI).

17. The computer program product of claim 16, further comprising: outputting a validation failed message in response to the similarity score being less than the predetermined threshold value; and outputting a validation successful message in response to the similarity score being greater than the predetermined threshold value, wherein the metric data comprises a central processing unit (CPU) utilization, memory utilization, and network bytes, and the determining the similarity score based on the first set of health scores and the second set of health scores occurs by using a time wrapped edit distance component between the first set of health scores and the second set of health scores.

18. The computer program product of claim 16, wherein the log data analysis module comprises: a textscalar which converts each log at a specific timestamp of the log data to a scalar value and multiplies the converted scalar value to create row vectors per file; a graph embedding component which receives the row vectors per file and uses a neural network (NN) algorithm to return links between log files based on their Euclidean distance; a graph attention network which receives

graph data comprising the links between the log files based on their Euclidean distance and generates node embedding vectors; a spatial short Fourier transform component which receives a matrix based on the row vector per file and converts the matrix into a spatial dimension and temporal dimension of a lower space; and a gated recurrent unit (GRU) which outputs a concatenated matrix which is a concatenation of the spatial dimension and temporal dimension of the lower space and another matrix based on a weightage of the log file.

19. The computer program product of claim 16, wherein the plurality of AI models comprise: a matrix profile model which receives the metric data and calculates a first set of anomaly points and anomaly scores using a matrix profile algorithm which computes distances for a windowed sub-sequence against an entire time series, sets an exclusion zone to ignore trivial matches, and updates a distance profile with minimal values, and sets a nearest-neighbor index; a STD model which receives the metric data and calculates a second set of anomaly points and anomaly scores using a STD algorithm which uses locally fitted regression models to decompose a time series into trend, seasonal, and remainder components; an elliptic envelop model which receives the metric data and calculates a third set of anomaly points and anomaly scores using an elliptic envelop algorithm which creates an elliptical area around a dataset of the metric data; and an isolation forest model which receives the metric data and calculates a fourth set of anomaly points and anomaly scores using an isolation forest algorithm which splits a data space randomly and uses a randomly selected attribute and a randomly selected split point to detect anomalies.

20. A system comprising: a processor, a computer readable memory, one or more computer readable storage media, and program instructions collectively stored on the one or more computer readable storage media, the program instructions executable to: receive metric and log data from an external system; generate a first set of health scores for metric data of the metric and log data using a plurality of artificial intelligence (AI) models which include a matrix profile model, a seasonal trend decomposition (STD) model, an elliptic envelop model, and an isolation forest model; generate a second set of health scores for log data of the metric and log data using a textscalar, a graph embedding component, a graph attention network, a spatial short Fourier transform component, and a gated recurrent unit (GRU); determine a similarity score based on the first set of health scores and the second set of health scores; perform a validation by comparing the similarity score to a predetermined threshold value; and output the first set of health scores and the second set of health scores and corresponding timestamps to a graphical user interface (GUI), wherein the metric data comprises a central processing unit (CPU) utilization, memory utilization, and network bytes.
